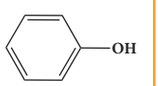
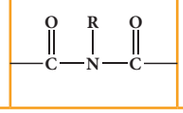
S. No.	Class of compounds	Functional group	General formula (R - Alkyl group)
16	Cyanide (Nitrile)	—C≡N	R-CN
17	Isocyanide	—NC	R-NC
18	Cyanate	—OCN	R-OCN
19	Isocyanate	—NCO	R-NCO
20	Thiocyanate	—SCN	R-SCN
21	Isothiocyanate	—NCS	R-NCS
23	Thioalcohols or thiols	—SH	R-SH
24	Thioethers	—S—R	R-S-R'
25	Imines	=NH	R-CH=NH
26	Nitroso compounds	—NO	R-NO
27	Hydrazines	—NH—NH_2	R-NH-NH_2
28	Hydrazo compounds	—NH—NH—	R-NH-NH-R
29	Phenols		C_6H_5OH
30	Imide		RCON(R') | COR''

11.3 Nomenclature of organic compounds:

The International Union of Pure and Applied Chemistry (IUPAC) is the world authority on chemical nomenclature and terminology, naming of new elements in the periodic table standardized methods for measurement; atomic weights, and many other critically-evaluated data. According to IUPAC recommendations to name any organic compound, it is considered as a derivative of its parent saturated hydrocarbon. The IUPAC name of an organic compound consists of three parts.

prefix + root word + suffix

Root word denotes the number of carbon atoms in the longest continiuous chain in molecules. Prefix denotes the group(s) attached to the main chain which is placed before the root. Suffix denotes the funtional group and is placed after the root word.

Table 11.2 Number of carbons in parent chain and the corresponding root words

Chain length(or) no. of carbon atoms	Root word	Chain length(or) no. of carbon atom	Root word
C_1	Meth-	C_{17}	Heptadec-
C_2	Eth-	C_{18}	Octadec-
C_3	Prop-	C_{19}	Nonadec-
C_4	But-	C_{20}	Icos-
C_5	Pent-	C_{21}	Henicos
C_6	Hex-	C_{22}	Docos
C_7	Hept-	C_{30}	Triacont-
C_8	Oct-	C_{31}	Hentriacont
C_9	Non-	C_{32}	Dotriacont
C_{10}	Dec-	C_{40}	Tetracont-
C_{11}	Undec-	C_{50}	Pentacont-

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